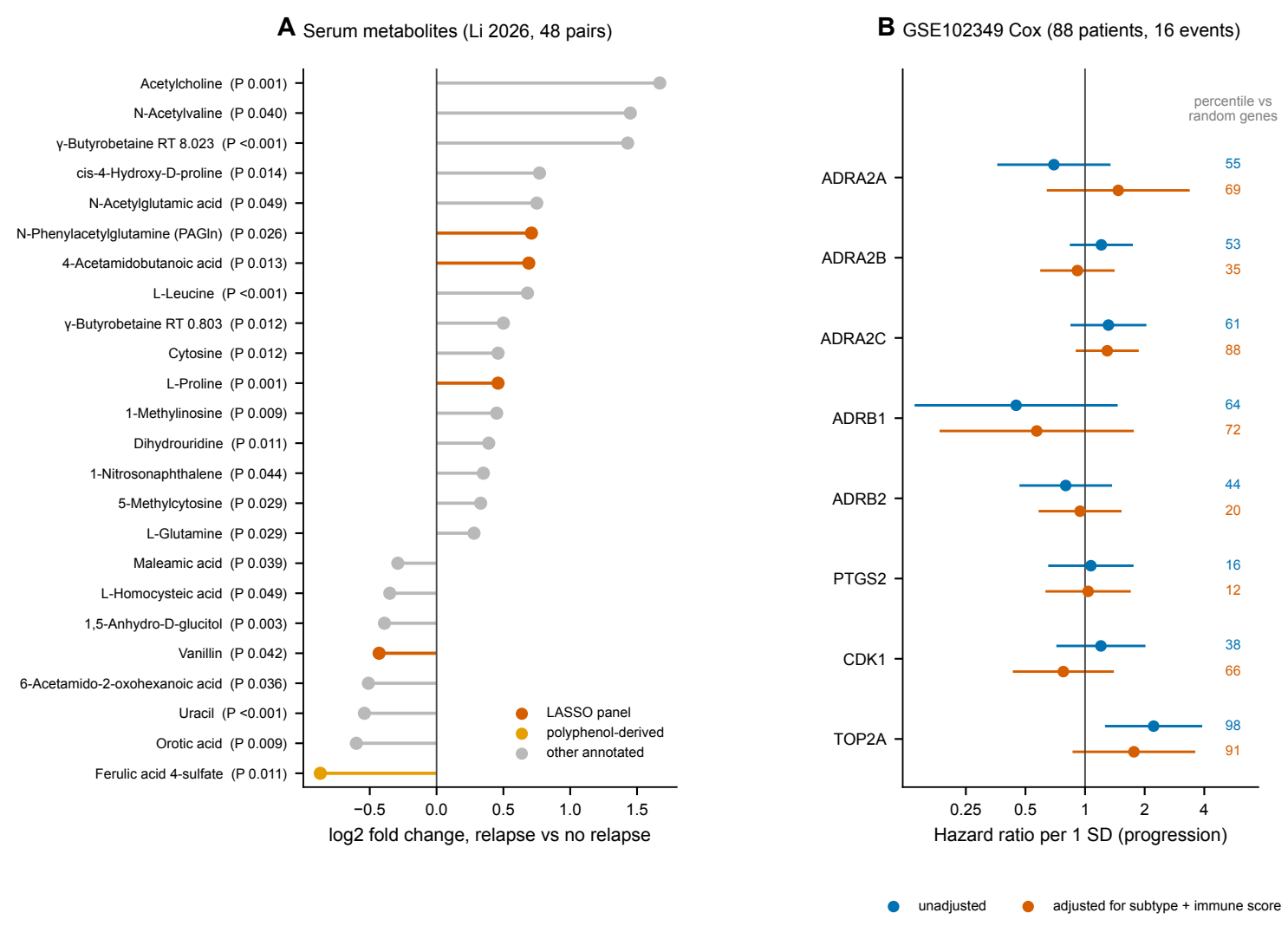

A: summary-level data; raw P values, no multiple-testing correction. Red = the four metabolites of the published LASSO-Cox recurrence panel.
B: eight genes fixed in advance: the five adrenergic receptors ADRA2A/2B/2C and ADRB1/2 (only ADRA2A, ADRA2B and ADRB2 are established PAGln receptors, Nemet 2020), plus PTGS2, CDK1 and TOP2A. The pre-specified criterion was an adjusted |z| above the 95th percentile of a null built from 2,000 random genes with mean FPKM > 1 (one shared null; not expression-matched).
No gene meets it after adjustment; TOP2A is the only gene above the threshold before adjustment (98th percentile, 91st after adjustment). The immune-score covariate has a similar effect size (HR 0.45 per SD, z = -2.80). 25 of 113 patients lacked event or follow-up time and were excluded.
Exploratory; no gene is claimed to predict outcome.